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BD_AIR_MAROC

A. CREATION DU TABLE :

Création de la table PILOTE :

```
mysql> create table PILOTE(  
  -> numpil varchar(10) not null primary key,  
  -> nompil varchar(50) not null,  
  -> ville varchar(20) not null,  
  -> salaire decimal(10,2) not null);  
Query OK, 0 rows affected (0.03 sec)
```

Création de la table AVION:

```
mysql> create table AVION(  
  -> numav varchar (10) not null primary key,  
  -> nomav varchar(50) not null,  
  -> capacite int(4) not null,  
  -> ville varchar(20) not null);  
Query OK, 0 rows affected, 1 warning (0.02 sec)
```

Création de la table VOL:

```
mysql> create table VOL(  
  -> numvol varchar(10) not null primary key,  
  -> numpil varchar(10) not null,  
  -> numav varchar(10) not null,  
  -> villedep varchar(10) not null,  
  -> villearr varchar(10) not null,  
  -> heuredep time not null,  
  -> heurearr time not null,  
  -> foreign key(numpil) references PILOTE(numpil),  
  -> foreign key(numav) references AVION(numav));  
Query OK, 0 rows affected (0.01 sec)
```

B. INSERTION DES DONNEES:

Dans la table PILOTE:

```
mysql> INSERT INTO PILOTE (numpil, nompil, ville, salaire) VALUES
-> (1, 'Sandrica', 'Marrakech', 25000.00),
-> (2, 'Alicia', 'Casablanca', 27000.00),
-> (3, 'Elisiah', 'Meknès', 18000.00),
-> (4, 'Majolie', 'Guelmim', 14000.00),
-> (5, 'Falina', 'Marrakech', 22000.00),
-> (6, 'Rojo', 'Casablanca', 15000.00),
-> (7, 'Kainael', 'Meknès', 29000.00),
-> (8, 'Kenzo', 'Guelmim', 12000.00),
-> (9, 'Lints', 'Marrakech', 17000.00),
-> (10, 'Angela', 'Casablanca', 20000.00);
Query OK, 10 rows affected (0.00 sec)
Enregistrements: 10  Doublons: 0  Avertissements: 0

mysql> SELECT * FROM PILOTE;
+-----+-----+-----+-----+
| numpil | nompil | ville   | salaire |
+-----+-----+-----+-----+
| 1      | Sandrica | Marrakech | 25000.00 |
| 2      | Alicia  | Casablanca | 27000.00 |
| 3      | Elisiah | Meknès   | 18000.00 |
| 4      | Majolie | Guelmim  | 14000.00 |
| 5      | Falina  | Marrakech | 22000.00 |
| 6      | Rojo    | Casablanca | 15000.00 |
| 7      | Kainael | Meknès   | 29000.00 |
| 8      | Kenzo   | Guelmim  | 12000.00 |
| 9      | Lints   | Marrakech | 17000.00 |
| 10     | Angela  | Casablanca | 20000.00 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

Dans la table AVION :

```
mysql> INSERT INTO AVION (numav, nomav, capacite, ville) VALUES
-> ('A001', 'Airbus A320', 180, 'Marrakech'),
-> ('A002', 'Boeing 737', 160, 'Casablanca'),
-> ('A003', 'Airbus A380', 500, 'Meknès'),
-> ('A004', 'Boeing 777', 350, 'Guelmim'),
-> ('A005', 'Embraer 190', 100, 'Marrakech'),
-> ('A006', 'Airbus A330', 280, 'Casablanca'),
-> ('A007', 'Boeing 787', 300, 'Meknès'),
-> ('A008', 'Airbus A350', 320, 'Guelmim'),
-> ('A009', 'Bombardier CRJ', 90, 'Marrakech'),
-> ('A010', 'Cessna 172', 4, 'Casablanca');
Query OK, 10 rows affected (0.00 sec)
Enregistrements: 10  Doublons: 0  Avertissements: 0

mysql> SELECT * FROM AVION;
+-----+-----+-----+-----+
| numav | nomav      | capacite | ville   |
+-----+-----+-----+-----+
| A001  | Airbus A320 | 180      | Marrakech |
| A002  | Boeing 737  | 160      | Casablanca |
| A003  | Airbus A380 | 500      | Meknès   |
| A004  | Boeing 777  | 350      | Guelmim  |
| A005  | Embraer 190 | 100      | Marrakech |
| A006  | Airbus A330 | 280      | Casablanca |
| A007  | Boeing 787  | 300      | Meknès   |
| A008  | Airbus A350 | 320      | Guelmim  |
| A009  | Bombardier CRJ | 90      | Marrakech |
| A010  | Cessna 172  | 4        | Casablanca |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

Dans la table VOL :

```
mysql> INSERT INTO VOL (numvol, numpil, numav, villedep, villearr, heuredep, heurearr) VALUES
-> ('V001', 1, 'A001', 'Marrakech', 'Casablanca', '08:00', '09:00'),
-> ('V002', 2, 'A002', 'Casablanca', 'Marrakech', '10:00', '11:00'),
-> ('V003', 3, 'A003', 'Meknès', 'Guelmim', '12:00', '14:30'),
-> ('V004', 4, 'A004', 'Guelmim', 'Marrakech', '09:00', '11:30'),
-> ('V005', 5, 'A005', 'Marrakech', 'Meknès', '13:00', '14:00'),
-> ('V006', 6, 'A006', 'Casablanca', 'Guelmim', '07:00', '09:00'),
-> ('V007', 7, 'A007', 'Meknès', 'Casablanca', '15:00', '16:30'),
-> ('V008', 8, 'A008', 'Guelmim', 'Marrakech', '17:00', '19:00'),
-> ('V009', 9, 'A009', 'Marrakech', 'Casablanca', '06:30', '07:30'),
-> ('V010', 10, 'A010', 'Casablanca', 'Meknès', '18:00', '19:30');
Query OK, 10 rows affected (0.00 sec)
Enregistrements: 10  Doublons: 0  Avertissements: 0

mysql> SELECT * FROM VOL;
+-----+-----+-----+-----+-----+-----+-----+
| numvol | numpil | numav | villedep | villearr | heuredep | heurearr |
+-----+-----+-----+-----+-----+-----+-----+
| V001   | 1      | A001  | Marrakech | Casablanca | 08:00:00 | 09:00:00 |
| V002   | 2      | A002  | Casablanca | Marrakech | 10:00:00 | 11:00:00 |
| V003   | 3      | A003  | Meknès    | Guelmim   | 12:00:00 | 14:30:00 |
| V004   | 4      | A004  | Guelmim   | Marrakech | 09:00:00 | 11:30:00 |
| V005   | 5      | A005  | Marrakech | Meknès    | 13:00:00 | 14:00:00 |
| V006   | 6      | A006  | Casablanca | Guelmim   | 07:00:00 | 09:00:00 |
| V007   | 7      | A007  | Meknès    | Casablanca | 15:00:00 | 16:30:00 |
| V008   | 8      | A008  | Guelmim   | Marrakech | 17:00:00 | 19:00:00 |
| V009   | 9      | A009  | Marrakech | Casablanca | 06:30:00 | 07:30:00 |
| V010   | 10     | A010  | Casablanca | Meknès    | 18:00:00 | 19:30:00 |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

C. REPONSES AUX QUESTIONS :

1.

```
mysql> SELECT *
-> FROM AVION
-> WHERE CAPACITE > 350;
+-----+-----+-----+-----+
| numav | nomav      | capacite | ville |
+-----+-----+-----+-----+
| A003   | Airbus A380 | 500      | Dubai |
+-----+-----+-----+-----+
1 row in set (0.02 sec)
```

2.

```
mysql> SELECT DISTINCT NUMPIL, VILLEDEP
-> FROM VOL;
+-----+-----+
| NUMPIL | VILLEDEP |
+-----+-----+
| 1      | Marrakech |
| 2      | Casablanca |
| 3      | Meknès    |
| 4      | Guelmim   |
| 5      | Marrakech |
| 6      | Casablanca |
| 7      | Meknès    |
| 8      | Guelmim   |
| 9      | Marrakech |
| 10     | Casablanca |
+-----+-----+
10 rows in set (0.02 sec)
```

3.

```
mysql> SELECT * FROM PILOTE;
+-----+-----+-----+-----+
| numpil | nompil | ville | salaire |
+-----+-----+-----+-----+
| 1      | Sandrica | Marrakech | 25000.00 |
| 2      | Alicia  | Casablanca | 27000.00 |
| 3      | Elisiah | Meknès   | 18000.00 |
| 4      | Majolie | Guelmim  | 14000.00 |
| 5      | Falina  | Marrakech | 22000.00 |
| 6      | Rojo    | Casablanca | 15000.00 |
| 7      | Kinael  | Meknès   | 29000.00 |
| 8      | Kenzo   | Guelmim  | 12000.00 |
| 9      | Lints   | Marrakech | 17000.00 |
| 10     | Angela  | Casablanca | 20000.00 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

4.

```
mysql> SELECT NOMPIL
-> FROM PILOTE
-> WHERE VILLE = 'Meknès'
-> AND SALAIRE > 20000;
+-----+
| NOMPIL |
+-----+
| Kinael |
+-----+
1 row in set (0.00 sec)
```

5.

```
mysql> SELECT NUMAV, NOMAV
-> FROM AVION
-> WHERE VILLE = 'Marrakech'
-> OR CAPACITE < 350;
+-----+-----+
| NUMAV | NOMAV |
+-----+-----+
| A001  | Airbus A320 |
| A002  | Boeing 737  |
| A005  | Embraer 190 |
| A006  | Airbus A330 |
| A007  | Boeing 787  |
| A008  | Airbus A350 |
| A009  | Bombardier CRJ |
| A010  | Cessna 172  |
+-----+-----+
8 rows in set (0.02 sec)
```

6.

```
mysql> SELECT NUMAV, NOMAV
-> FROM AVION
-> WHERE VILLE = 'Marrakech'
-> OR CAPACITE < 350;
```

NUMAV	NOMAV
A001	Airbus A320
A002	Boeing 737
A005	Embraer 190
A006	Airbus A330
A007	Boeing 787
A008	Airbus A350
A009	Bombardier CRJ
A010	Cessna 172

8 rows in set (0.02 sec)

7.

```
mysql> SELECT NUMPIL
-> FROM PILOTE
-> WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);
Empty set (0.02 sec)
```

8.

```
mysql> SELECT V.NUMVOL
-> FROM VOL V, PILOTE P
-> WHERE V.NUMPIL = P.NUMPIL
-> AND V.VILLEDEP = 'Marrakech'
-> AND P.VILLE = 'Meknès';
Empty set (0.00 sec)
```

9.

```
mysql> SELECT *
-> FROM VOL V, AVION A
-> WHERE V.NUMAV = A.NUMAV
-> AND A.VILLE <> 'Marrakech';
```

numvol	numpil	numav	villedep	villearr	heuredep	heurearr	numav	nomav	capacite	ville
V002	2	A002	Casablanca	Marrakech	10:00:00	11:00:00	A002	Boeing 737	160	Casablanca
V003	3	A003	Meknès	Guelmim	12:00:00	14:30:00	A003	Airbus A380	500	Meknès
V004	4	A004	Guelmim	Marrakech	09:00:00	11:30:00	A004	Boeing 777	350	Guelmim
V006	6	A006	Casablanca	Guelmim	07:00:00	09:00:00	A006	Airbus A330	280	Casablanca
V007	7	A007	Meknès	Casablanca	15:00:00	16:30:00	A007	Boeing 787	300	Meknès
V008	8	A008	Guelmim	Marrakech	17:00:00	19:00:00	A008	Airbus A350	320	Guelmim
V010	10	A010	Casablanca	Meknès	18:00:00	19:30:00	A010	Cessna 172	4	Casablanca

7 rows in set (0.00 sec)

10.

```
mysql> SELECT VILLEARR
-> FROM VOL
-> WHERE VILLEDEP = 'Guelmin';
Empty set (0.00 sec)
```

11.

```
mysql> SELECT P.NOMPIL, V.NUMVOL
-> FROM PILOTE P, VOL V
-> WHERE P.NUMPIL = V.NUMPIL;
+-----+-----+
| NOMPIL | NUMVOL |
+-----+-----+
| Sandrica | V001 |
| Alicia | V002 |
| Elisiah | V003 |
| Majolie | V004 |
| Falina | V005 |
| Rojo | V006 |
| Kinael | V007 |
| Kenzo | V008 |
| Lints | V009 |
| Angela | V010 |
+-----+-----+
10 rows in set (0.00 sec)
```

12.

```
mysql> SELECT P.NOMPIL, A.NOMAV
-> FROM PILOTE P, VOL V, AVION A
-> WHERE P.NUMPIL = V.NUMPIL
-> AND A.NUMAV = V.NUMAV;
+-----+-----+
| NOMPIL | NOMAV |
+-----+-----+
| Sandrica | Airbus A320 |
| Alicia | Boeing 737 |
| Elisiah | Airbus A380 |
| Majolie | Boeing 777 |
| Falina | Embraer 190 |
| Rojo | Airbus A330 |
| Kinael | Boeing 787 |
| Kenzo | Airbus A350 |
| Lints | Bombardier CRJ |
| Angela | Cessna 172 |
+-----+-----+
10 rows in set (0.00 sec)
```

13.

```
mysql> SELECT V.NUMVOL, V.VILLEDEP, V.VILLEARR, P.NOMPIL  
-> FROM VOL V, PILOTE P  
-> WHERE V.NUMPIL = P.NUMPIL;
```

NUMVOL	VILLEDEP	VILLEARR	NOMPIL
V001	Marrakech	Casablanca	Sandrica
V002	Casablanca	Marrakech	Alicia
V003	Meknès	Guelmim	Elisiah
V004	Guelmim	Marrakech	Majolie
V005	Marrakech	Meknès	Falina
V006	Casablanca	Guelmim	Rojo
V007	Meknès	Casablanca	Kainael
V008	Guelmim	Marrakech	Kenzo
V009	Marrakech	Casablanca	Lints
V010	Casablanca	Meknès	Angela

10 rows in set (0.00 sec)

14.

```
mysql> SELECT DISTINCT A.NOMAV  
-> FROM VOL V, AVION A  
-> WHERE V.NUMAV = A.NUMAV  
-> AND V.VILLEDEP = 'Casablanca';
```

NOMAV
Boeing 737
Airbus A330
Cessna 172

3 rows in set (0.00 sec)

15.

```
mysql> SELECT DISTINCT P.NOMPIL  
-> FROM PILOTE P, VOL V, AVION A  
-> WHERE P.NUMPIL = V.NUMPIL  
-> AND V.NUMAV = A.NUMAV  
-> AND A.CAPACITE > 300;
```

NOMPIL
Elisiah
Majolie
Kenzo

3 rows in set (0.00 sec)

16.

```
mysql> SELECT P.NOMPIL, V.VILLEARR
-> FROM PILOTE P, VOL V
-> WHERE P.NUMPIL = V.NUMPIL;
```

NOMPIL	VILLEARR
Sandrica	Casablanca
Alicia	Marrakech
Elisiah	Guelmim
Majolie	Marrakech
Falina	Meknès
Rojo	Guelmim
Kainael	Casablanca
Kenzo	Marrakech
Lints	Casablanca
Angela	Meknès

10 rows in set (0.00 sec)

17.

```
mysql> SELECT *
-> FROM VOL V
-> JOIN PILOTE P ON V.NUMPIL = P.NUMPIL
-> JOIN AVION A ON V.NUMAV = A.NUMAV;
```

numvol	numpil	numav	villedep	villearr	heuredep	heurearr	numpil	nompil	ville	salaire	numav	nomav	capacite	ville
V001	1	A001	Marrakech	Casablanca	08:00:00	09:00:00	1	Sandrica	Marrakech	25000.00	A001	Airbus A320	180	Marrakech
V002	2	A002	Casablanca	Marrakech	10:00:00	11:00:00	2	Alicia	Casablanca	27000.00	A002	Boeing 737	160	Casablanca
V003	3	A003	Meknès	Guelmim	12:00:00	14:30:00	3	Elisiah	Meknès	18000.00	A003	Airbus A380	500	Meknès
V004	4	A004	Guelmim	Marrakech	09:00:00	11:30:00	4	Majolie	Guelmim	14000.00	A004	Boeing 777	350	Guelmim
V005	5	A005	Marrakech	Meknès	13:00:00	14:00:00	5	Falina	Marrakech	22000.00	A005	Embraer 190	100	Marrakech
V006	6	A006	Casablanca	Guelmim	07:00:00	09:00:00	6	Rojo	Casablanca	15000.00	A006	Airbus A330	280	Casablanca
V007	7	A007	Meknès	Casablanca	15:00:00	16:30:00	7	Kainael	Meknès	29000.00	A007	Boeing 787	300	Meknès
V008	8	A008	Guelmim	Marrakech	17:00:00	19:00:00	8	Kenzo	Guelmim	12000.00	A008	Airbus A350	320	Guelmim
V009	9	A009	Marrakech	Casablanca	06:30:00	07:30:00	9	Lints	Marrakech	17000.00	A009	Bombardier CRJ	90	Marrakech
V010	10	A010	Casablanca	Meknès	18:00:00	19:30:00	10	Angela	Casablanca	20000.00	A010	Cessna 172	4	Casablanca

10 rows in set (0.00 sec)

18.

```
mysql> SELECT *
-> FROM PILOTE
-> WHERE SALAIRE > (SELECT AVG(SALAIRE) FROM PILOTE);
```

numpil	nompil	ville	salaire
1	Sandrica	Marrakech	25000.00
2	Alicia	Casablanca	27000.00
5	Falina	Marrakech	22000.00
7	Kainael	Meknès	29000.00
10	Angela	Casablanca	20000.00

5 rows in set (0.03 sec)

19.

```
mysql> SELECT *
-> FROM AVION
-> WHERE CAPACITE > (SELECT CAPACITE FROM AVION WHERE NUMAV = '101');
```

Empty set (0.02 sec)

20.

```
mysql> SELECT *  
  -> FROM PILOTE  
  -> WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);  
Empty set (0.00 sec)
```

21.

```
mysql> SELECT DISTINCT V.VILLEDEP  
  -> FROM VOL V, PILOTE P  
  -> WHERE V.NUMPIL = P.NUMPIL  
  -> AND P.VILLE = 'Marrakech';  
+-----+  
| VILLEDEP |  
+-----+  
| Marrakech |  
+-----+  
1 row in set (0.00 sec)
```

22.

```
mysql> CREATE VIEW VUE_VOL AS  
  -> SELECT V.NUMVOL, P.NOMPIL, A.NOMAV, V.VILLEDEP, V.VILLEARR  
  -> FROM VOL V  
  -> JOIN PILOTE P ON V.NUMPIL = P.NUMPIL  
  -> JOIN AVION A ON V.NUMAV = A.NUMAV;  
Query OK, 0 rows affected (0.01 sec)
```

23.

```
mysql> CREATE VIEW VUE_PILOTE AS  
  -> SELECT *  
  -> FROM PILOTE  
  -> WHERE SALAIRE > 15000;  
Query OK, 0 rows affected (0.02 sec)
```

24.

```
mysql> SELECT * FROM VUE_PILOTE;  
+-----+-----+-----+-----+  
| numpil | nompil | ville   | salaire |  
+-----+-----+-----+-----+  
| 1      | Sandrica | Marrakech | 25000.00 |  
| 2      | Alicia  | Casablanca | 27000.00 |  
| 3      | Elisiah | Meknès   | 18000.00 |  
| 5      | Falina  | Marrakech | 22000.00 |  
| 7      | Kainael | Meknès   | 29000.00 |  
| 9      | Lints   | Marrakech | 17000.00 |  
| 10     | Angela  | Casablanca | 20000.00 |  
+-----+-----+-----+-----+  
7 rows in set (0.02 sec)
```

25.

```
mysql> DROP VIEW VUE_PILOTE;  
Query OK, 0 rows affected (0.02 sec)
```

26.

```
mysql> CREATE VIEW VUE_VOL_MAR AS  
-> SELECT *  
-> FROM VOL  
-> WHERE VILLEDEP = 'Marrakech';  
Query OK, 0 rows affected (0.02 sec)
```

27.

```
mysql> CREATE USER 'user_air'@'localhost' IDENTIFIED BY '1234';  
Query OK, 0 rows affected (0.05 sec)
```

28.

```
mysql> GRANT SELECT ON PILOTE TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.00 sec)
```

29.

```
mysql> GRANT INSERT ON VOL TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.02 sec)
```

30.

```
mysql> GRANT DELETE ON AVION TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> REVOKE DELETE ON AVION FROM 'user_air'@'localhost';  
Query OK, 0 rows affected (0.00 sec)
```

31.

```
mysql> GRANT ALL PRIVILEGES ON BD_AIR_MAROC.* TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.02 sec)
```

32.

```
mysql> DROP ROLE IF EXISTS gestion_vol;  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> CREATE ROLE gestion_vol;  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> GRANT SELECT, INSERT, UPDATE, DELETE ON VOL TO gestion_vol;  
Query OK, 0 rows affected (0.02 sec)
```

33.

```
mysql> GRANT gestion_vol TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.00 sec)
```